Now, let us learn subtraction of integers in another way.

Observe the following patterns :

$7-2=5$; $7-1=6$; $7-0=7$

What will happen if we extend this to negative integers?

$7-(-1)=8$; $7-(-2)=9$; $7-(-3)=10$

We shall see some more patterns.

$20-2=18$; $20-1=19$; $20-0=20$; $20-(-1)=21$; $20-(-2)=22$

We can see from the above patterns that while subtracting consecutive negative integers from 7 and 20 the difference increase consecutively.

It is clear that subtraction of negative integers gives increase in the difference. For example $7-(-2)=9$. Hence, subtraction of -2 is equivalent to addition of 2, which is the additive inverse of -2. That is, 7+2 =9.

So, to subtract a negative integer from an integer we add the additive inverse of the integer which is to be subtracted.

For example, subtract (–5) from 7.

7 – (–5)

To subtract (-5) we can add additive inverse of (–5) that is 5 with 7

Therefore, 7 – (–5) = 12

Try these

1. Do the following by using number line.
   (i) (–4) – (+3) (ii) (–4) – (–3)
2. Find the values and compare the answers.
   (i) (–6) – (–2) and (–6) + 2
   (ii) 35 – (–7) and 35 + 7
   (iii) 26 – (+10) and 26 + (–10)
3. Put the suitable symbol <, > or = in the boxes.
   (i) −10 − 8 ☐ −10 +8 (ii) (−20) + 10 ☐ (−20) − (−10)
   (iii) (−70) − (−50) ☐ (−70) −50 (iv) 100 − (+100) ☐ 100 − (−100)
   (v) −50 −30 ☐ −100 + 20

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